PROPORTOIDS

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Abstract. Analogical proportions are expressions of the form "a is to b what c is to d" at the core of analogical reasoning. This paper contributes to the mathematical foundations of analogical proportions in the axiomatic tradition as initiated — in the tradition of the ancient Greeks — by Yves Lepage two decades ago. More precisely, we first introduce the name "proportoid" for sets endowed with a 4-ary analogical proportion relation satisfying a suitable set of axioms. We then study study different kinds of proportion-preserving mappings and relations and their properties. Formally, we define homomorphisms of proportoids as mappings H satisfying $a : b :: c : d$ iff $\mathsf{H}a : \mathsf{H}b :: \mathsf{H}c : \mathsf{H}d$ for all elements and show that their kernel is a congruence. Moreover, we introduce (proportional) analogies as mappings A satisfying $a : b :: \mathsf{A}a : \mathsf{A}b$ for all elements a and b in the source domain and show how to compute partial analogies. We then introduce a number of useful relations between functions (including homomorphisms and analogies) on proportoids and study their properties. In a broader sense, this paper is a further step towards a mathematical theory of analogical proportions.

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1. Introduction

Analogical proportions are expressions of the form "a is to b what c is to d" at the core of analogical reasoning [10].

The purpose of this paper is to further develop from a mathematical point of view the axiomatic approach to analogical proportions as initiated — in the tradition of the ancient Greeks — by Lepage [13] two decades ago. We will mostly remain agnostic regarding the concrete set of assumed axioms as we will only assume three evident properties.

More formally, we define ***proportoids*** §3 as sets endowed with a 4-ary analogical proportion relation $a : b :: c : d$ satisfying the axioms (1)–(3) (cf. Definition 2), and we define ***subproportoids***, ***homomorphisms*** §4, and ***congruences*** §5. It is important to emphasize that all constructions in this paper are kept very general and do not refer to a concrete instance of an analogical proportion relation.[1]

Functions preserving analogical proportions have already proven to be of practical interest (e.g. [6]), and studying their mathematical properties is essential for understanding proportions. We therefore define the notion of a ***proportional homomorphism*** §4 preserving the analogical proportion relation across different domains by satisfying a stronger version of the proportional inference principle of [6] which corresponds to the analogical jump in [8] (cf. [7]); cf. Remark 11. In §5, we then introduce the fundamental concept of a congruence on a proportoid and show in Theorem 21 that the ***kernel*** of every homomorphism is a congruence as expected.

In a similar vein, we then introduce the notion of a ***proportional analogy*** §7 preserving the relationship between elements of the source domain. We show in Theorem 34 that analogies satisfy the ***strong proportion-preserving property*** (cf. Remark 11) given that the underlying proportoids are transitive. In §8, we provide a procedure for constructing ***partial proportional analogies*** using an enumeration of the source structure and a selection function.

In §9, we define the notion of a ***proportional identity*** and show that it is a congruence relation in a wide class of proportoids (Theorem 60).

Functions between proportoids are fundamental. In §10, we therefore introduce a number of binary relations between functions on proportoids which we believe are appealing from a mathematical point of view since most of them are equivalence relations and some are congruences.

Finally, in §13 we introduce a notion of ***similarity*** using analogical proportions and show in Theorem 122 that it is a congruence under mild conditions.

In a broader sense, this paper is a further step towards a mathematical theory of analogical proportions.

2. Preliminaries

A set P is ***denumerable*** iff there is a surjective mapping $\mathbb{N} \to P$.

[1] However, we do want to point out the algebro-logical framework in [1, 4] which provides a *canonical* analogical proportion relation in *any* algebraic and logical structure containing functions and relations (see §14).

We denote the ***identity function*** on any set by I. A function F on a partially ordered set $(P, \leq)$ is ***monotone*** iff $a \leq \mathsf{F}a$ holds for every $a \in P$. A ***partial function*** from P to R is a function $\mathsf{F} : P \to R \cup \{\mathbf{u}\}$ possibly yielding the value $\mathbf{u}$ ("undefined") for some elements of P. We define the ***domain*** of such a function by

$$dom\,\mathsf{F} := \{a \in P \mid \mathsf{F}a \neq \mathbf{u}\}.$$

We call F ***total*** iff $dom\,\mathsf{F} = P$.

Let ρ be a binary relation on unary mappings on P. An unary mapping F is ***idempotent*** with respect to ρ iff $\mathsf{F}\rho\mathsf{F}^2$. As usual, the relation ρ is ***left compatible*** iff

$$\mathsf{F}\rho\mathsf{G} \quad \Rightarrow \quad \mathsf{EF}\rho\mathsf{EG}, \quad \text{for all } \mathsf{E},$$

right compatible iff

$$\mathsf{F}\rho\mathsf{G} \quad \Rightarrow \quad \mathsf{FE}\rho\mathsf{GE}, \quad \text{for all } \mathsf{E},$$

and ***compatible*** iff

$$\mathsf{E}\rho\mathsf{F} \quad \text{and} \quad \mathsf{G}\rho\mathsf{H} \quad \Rightarrow \quad \mathsf{EG}\rho\mathsf{FH}.$$

A left [right] compatible equivalence is called a ***left*** [***right***] ***congruence***. A ***congruence*** is a compatible equivalence relation.

The following characterization of congruences is folklore (see e.g. [11, Proposition 1.5.1]):

Proposition 1. *A binary relation on a semigroup is a congruence iff it is both a left and a right congruence.*

3. Proportoids

In this section, we formally introduce proportoids as sets endowed with a 4-ary analogical proportion relation satisfying a minimal suitable set of axioms in the axiomatic tradition initiated by Yves Lepage [13] two decades ago:

Definition 2. A ***proportoid*** is a pair $\mathfrak{P} = (P, ::)$ consisting of a non-empty set P endowed with a 4-ary analogical proportion relation :: on P satisfying the following axioms, for all $a, b, c, d \in P$:

(1) $a : b :: a : b$ (reflexivity; r),

(2) $a : b :: c : d \quad \Leftrightarrow \quad c : d :: a : b$ (symmetry; s),

(3) $a : b :: c : d \quad \Leftrightarrow \quad b : a :: d : c$ (inner symmetry; y).

Beyond that, we will consider the following properties, for $a, b, c, d, e, f \in P$ and injective function $f : P \to P$:

(4) $a : a :: c : c$ (inner reflexivity; e),

(5) $a : b :: c : d \quad \Leftrightarrow \quad a : c :: b : d$ (central permutation; c),

(6) $a : a :: a : d \quad \Leftrightarrow \quad d = a$ (determinism; d),

(7) $a : b :: b : a$ (commutativity; o),

(8) $a : b :: c : d$ and $c : d :: e : f \quad \Rightarrow \quad a : b :: e : f$ (transitivity; t),

(9) $a : b :: c : d$ and $b : e :: d : f \quad \Rightarrow \quad a : e :: c : f$ (inner transitivity; i),

(10) $a : a :: c : d \quad \Rightarrow \quad d = c$ (strong inner reflexivity; v),

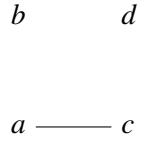
(11) $a : b :: a : d \quad \Rightarrow \quad d = b$ (strong reflexivity; x),

(12) $a : fa :: c : fc$, (functionality; f).

The symbol after the semicolon like e.g. "t" will be used in proofs as shortcuts for the respective axiom (in this case "transitivity") and to denote proportoids satisfying the axiom — for example, a proportoid satisfying transitivity is called a ***t-proportoid*** etc. Moreover, define the ***solution set*** of the ***proportional equation*** (or ***p-equation***) $a : b :: c : x$ by

$$\mathcal{S}_{\mathfrak{P}}(a : b :: c : x) := \{d \in P \mid a : b :: c : d\}.$$

Remark 3. Lepage [13] uses the axioms (2), (5), (10), and (11) in his axiomatization in the linguistic context.[2] We agree with (2). Although often accepted, central permutation (5) is debatable as shown by the simple counterexample:



Here we cannot say that a is to c as b is to d since there is a relation between a and c and none between b and d. That central permutation is problematic has been observed by others as well (see e.g. [14, 15]). The remaining axioms in Lepage's original list are in general inadequate as well as demonstrated by simple counterexamples as in the proof of [1, Theorem 28].

[2] Lepage [13] used other names for the axioms — we adapt here the terminology in [1, §4.3].

Example 4. Let $\leq$ be a partial ordering on P. Define

$$a : b ::_{\leq} c : d \quad :\Leftrightarrow \quad (a = b \quad \text{and} \quad c = d) \quad \text{or} \quad (a < b \quad \text{and} \quad c < d) \quad \text{or} \quad (a > b \quad \text{and} \quad d > c).$$

Then $(P, ::_{\leq})$ is a proportoid.

Example 5. Let θ be a reflexive and symmetric binary relation on P. Define

$$a : b ::_{\theta} c : d \quad :\Leftrightarrow \quad a\theta b \quad \text{and} \quad c\theta d.$$

Then $(P, ::_{\theta})$ is a proportoid. If θ is transitive, then $(P, ::_{\theta})$ is transitive.

Example 6. In $\mathbb{N}$, define the well-known ***difference proportion relation*** (and see [2, Difference Proportion Theorem]) as follows:

$$a : b :: c : d \quad :\Leftrightarrow \quad a - b = c - d.$$

Then $(\mathbb{N}, ::)$ is a proportoid satisfying *all* properties in Definition 2.

Example 7. A ***metric space*** is a set M together with a ***distance function*** $\delta : M \times M \to \mathbb{R}$ satisyfing, for all $a, b, c \in M$,

$$\begin{aligned}
&\delta(a, a) = 0,\\
&\delta(a, b) > 0 \quad \text{whenever} \quad a \neq b \quad \textbf{\textit{(positivity)}},\\
&\delta(a, b) = \delta(b, a) \quad \textbf{\textit{(symmetry)}},\\
&\delta(a, c) \leq \delta(a, b) + \delta(b, c) \quad \textbf{\textit{(triangle inequality)}}.
\end{aligned}$$

We endow every metric space (M, δ) with the 4-ary analogical proportion relation defined, for all $a, b, c, d \in M$, by

$$a : b ::_{\delta} c : d \quad :\Leftrightarrow \quad \delta(a, b) = \delta(c, d). \tag{13}$$

Then $(M, ::_{\delta})$ is a proportoid.

Example 8. Let $\circ$ be a binary operation on P. Define

$$a : b ::_{\circ} c : d \quad :\Leftrightarrow \quad a \circ b = c \circ d.$$

Then $(P, ::_{\circ})$ is *not* necessarily a proportoid. To see why, consider the operation $\circ$ defined, for all $a, b \in P$, by

$$a \circ b := b.$$

Let $c \in P$ be different from a. Then $a \circ b = c \circ b$, and therefore $a : b :: c : b$. Then inner symmetry makes that $b : a :: b :$ should hold. However, the latter fails to be the case because $b \circ a = a$ and $b \circ c = c$ and a and c are different by hypothesis.

With a notion of structure there is always an associated notion of substructure:

Definition 9. A proportoid $\mathfrak{R} = (R, ::)$ is a ***subproportoid*** of $\mathfrak{P} = (P, ::)$ iff $R \subseteq P$ and $a : b :: c : d$ holds in $\mathfrak{R}$ iff it holds in $\mathfrak{P}$, for all $a, b, c, d \in R$.

Intuitively, a subproportoid preserves the analogical proportion relation between the elements which in general may not be the case. For example, it may be the case that $a : b :: c : d$ holds in the proportoid $\mathfrak{R}$, whereas if we consider the larger proportoid $\mathfrak{P}$ we can find some d' such that the relation between a and b and between c and d' is more similar than between c and d.

4. Proportional homomorphisms

With any kind of structure there comes a notion of a structure-preserving mapping and proportoids are no exception. In the rest of the paper, let $\mathfrak{P} = (P, ::_{\mathfrak{P}})$ and $\mathfrak{R} = (R, ::_{\mathfrak{R}})$ be proportoids where we often omit the indices from notation in case the underlying proportoids are clear from the context.

Definition 10. We call a mapping $\mathsf{H} : P \to R$ a ***(proportional) homomorphism*** iff for all $a, b, c, d \in P$,

$$a : b ::_{\mathfrak{P}} c : d \quad \Leftrightarrow \quad \mathsf{H}a : \mathsf{H}b ::_{\mathfrak{R}} \mathsf{H}c : \mathsf{H}d. \tag{14}$$

A ***(proportional) endomorphism*** is a homomorphism of the form $\mathfrak{P} \to \mathfrak{P}$, and a ***(proportional) epimorphism*** is an onto homomorphism. A ***(proportional) isomorphism*** is a bijective homomorphism. We call $\mathfrak{P}$ and $\mathfrak{R}$ are ***(proportionally) isomorphic*** — in symbols, $\mathfrak{P} \cong \mathfrak{R}$ — iff there is a proportional isomorphism from $\mathfrak{P}$ to $\mathfrak{R}$.

Remark 11. The only if part "$\Rightarrow$" of the equivalence in (14) is called the *analogical inference principle* by [6] (and see [7]) and it can be viewed as a particular case of the so-called *analogical jump* by [8]. We prefer using the term "proportional" instead of "analogical" and ***proportion-preserving property*** (or ***PPP***) instead of "analogical inference principle", and we call the equivalence in (14) the ***strong proportion-preserving property*** (or ***sPPP***).

Remark 11 motivates the following definition:

Definition 12. A ***proportion-preserving mapping*** (or ***pp-mapping***) is any function $\mathsf{P} : \mathfrak{P} \to \mathfrak{R}$ satisfying

$$a : b ::_{\mathfrak{P}} c : d \quad \Rightarrow \quad \mathsf{P}a : \mathsf{P}b ::_{\mathfrak{R}} \mathsf{P}c : \mathsf{P}d, \quad \text{for all } a, b, c, d \in P. \tag{15}$$

Example 13. The ***k-successor function***, $k \geq 0$, given by

$$S^k a := a + k,$$

for all $a \in \mathbb{N}$, is a homomorphism on $(\mathbb{N}, ::)$, which is defined as in Example 6.

Example 14. Define the ***boolean proportion relation*** in $\mathfrak{R} := (\{0,1\}, ::)$ as follows [3, 12]:

$$a : b :: c : d \quad :\Leftrightarrow \quad (a = b \quad \text{and} \quad c = d) \quad \text{or} \quad (a \neq b \quad \text{and} \quad c \neq d).$$

The negation function $\neg$ is an isomorphism since it is bijective and satisfies

$$\begin{aligned} a : b :: c : d \quad &\Leftrightarrow \quad (a = b \quad \text{and} \quad c = d) \quad \text{or} \quad (a \neq b \quad \text{and} \quad c \neq d) \\ &\Leftrightarrow \quad (\neg a = \neg b \quad \text{and} \quad \neg c = \neg d) \quad \text{or} \quad (\neg a \neq \neg b \quad \text{and} \quad \neg c \neq \neg d) \\ &\Leftrightarrow \quad \neg a : \neg b :: \neg c : \neg d. \end{aligned}$$

Proposition 15. *The space of all endomorphisms forms a monoid with respect to function composition with the neutral element given by the identity function.*

Proof. It follows from the definition that homomorphisms are closed under composition, that is, if H and G are homomorphisms, then $\mathsf{H} \circ G$ is a homomorphism as well:

$$\begin{aligned} a : b :: c : d \quad &\Leftrightarrow \quad \mathsf{H}a : \mathsf{H}b :: \mathsf{H}c : \mathsf{H}d \\ &\Leftrightarrow \quad \mathsf{GH}a : \mathsf{GH}b :: \mathsf{GH}c : \mathsf{GH}d. \end{aligned}$$

The identity function is clearly a homomorphism. □

Theorem 16 (First Injectivity Theorem)**.** *Every homomorphism defined on a d-proportoid is injective.*

Proof. The following derivation shows that $\mathsf{H}a = \mathsf{H}b$ implies $a = b$, for any homomorphism $\mathsf{H} : \mathfrak{P} \to \mathfrak{R}$ on a d-proportoid $\mathfrak{P}$ and $a, b \in P$:

$$\dfrac{\dfrac{\dfrac{\mathsf{H}a = \mathsf{H}b}{\mathsf{H}a : \mathsf{H}a :: \mathsf{H}a : \mathsf{H}b}\,\mathrm{r}}{a : a :: a : b}\,\text{sPPP 11}}{a = b.}\,\mathrm{d}$$

□

Remark 17. Theorem 16 motivates the study of non-strong proportion-preserving functions as in [6] (cf. Remark 11).

5. Proportional congruences

In universal algebra, congruences provide a mechanism for factorizing algebras into equivalence classes compatible with the algebraic operations. Here, we require that the equivalence classes preserve the analogical proportion relation giving rise to the notion of a proportional congruence:[3]

Definition 18. An equivalence relation θ on P is a ***(proportional) congruence*** on $\mathfrak{P} = (P, ::)$ iff for all elements $a, b, c, d, e, f, g, h \in P$,

$$\frac{a\theta e \quad b\theta f \quad c\theta g \quad d\theta h \qquad a : b :: c : d}{e : f :: g : h}$$

or, equivalently,

$$\frac{a\theta e \quad b\theta f \quad c\theta g \quad d\theta h}{a : b :: c : d \quad \Leftrightarrow \quad e : f :: g : h.}$$

Proposition 19. *For any congruence θ on $\mathfrak{P}$,*

$$\frac{a\theta c \qquad b\theta d}{a : b :: c : d}\,.$$

Proof. The following inference rule is an instance of the rule defining a congruence in Definition 18:

$$\frac{a\theta a \quad b\theta b \quad a\theta c \quad b\theta d \qquad a : b :: a : b}{a : b :: c : d.}$$

Now observe that since θ is reflexive and analogical proportions are reflexive, we can omit $a\theta a$, $b\theta b$, and $a : b :: a : b$ in the first line of the rule which immediately yields the first implication of the proposition. □

A standard construction in universal algebra is given by the kernel of a homomorphism (cf. [5, Definition 6.7]) which we directly adapt here:

[3] We will show in Theorem 21 that proportional congruences and homomorphisms are connected via kernels.

Definition 20. The ***kernel*** of a homomorphism $\mathsf{H} : \mathfrak{P} \to \mathfrak{R}$ is given by

$$ker\ \mathsf{H} := \left\{ab \in P^2 \;\middle|\; \mathsf{H}a = \mathsf{H}b\right\}.$$

We now show that homomorphisms and congruences of proportoids are in the same way related as in universal algebra via kernels:

Theorem 21. *The kernel of any homomorphism is a congruence.*

Proof. Let $\mathsf{H} : \mathfrak{P} \to \mathfrak{R}$ be a homomorphism. We show

$$\frac{\mathsf{H}a = He \quad \mathsf{H}b = \mathsf{H}f \quad \mathsf{H}c = \mathsf{H}g \quad \mathsf{H}d = \mathsf{H}h \qquad a : b :: c : d}{e : f :: g : h}$$

for all $a, b, c, d, e, f, g, h \in P$, by the following derivation:

$$\dfrac{\dfrac{\mathsf{H}a = He \quad \mathsf{H}b = \mathsf{H}f \quad \mathsf{H}c = \mathsf{H}g \quad \mathsf{H}d = \mathsf{H}h \qquad \dfrac{a : b :: c : d}{\mathsf{H}a : \mathsf{H}b :: \mathsf{H}c : \mathsf{H}d}\ \text{PPP 11}}{\dfrac{\mathsf{H}e : \mathsf{H}f :: \mathsf{H}g : \mathsf{H}h}{e : f :: g : h.}\ \text{sPPP 11}}}{}$$

□

6. Proportional polymorphisms

The following definition follows the standard definition of a polymorphism on a relational proportoid:

Definition 22. A ***(proportional) polymorphism*** of $\mathfrak{P} = (P, ::)$ is any operation $f : P^n \to P$ preserving the analogical proportion relation in the sense that

$$\frac{a_1 : b_1 :: c_1 : d_1 \qquad \ldots \qquad a_n : b_n :: c_n : d_n}{fa_1 \ldots a_n : fb_1 \ldots b_n :: fc_1 \ldots c_n : fd_1 \ldots d_n}\ \text{PP}$$

holds for all $a_i, b_i, c_i, d_i \in P$, $1 \leq i \leq n$.

Remark 23. Notice that for any unary function $f : P \to P$, the above condition amounts to the proportion-preserving property (cf. Remark 11)

$$\frac{a : b :: c : d}{fa : fb :: fc : fd}\ \text{PPP}.$$

Definition 24. A ***strong (proportional) polymorphism*** (or ***s-polymorphism***) is a polymorphism where the implication of the inference rule (PP) is turned into an equivalence.

Example 25. The iterated successor function S^k is an s-polymorphism of $(\mathbb{N}, ::)$ defined as in Example 6, thus satisfying

$$a : b :: c : d \quad \Leftrightarrow \quad S^k a : S^k b :: S^k c : S^k d, \quad \text{for all } a, b, c, d \in \mathbb{N} \text{ and } k \geq 0.$$

Moreover, addition is a polymorphism of $(\mathbb{N}, ::)$ thus satisfying

$$\frac{a : b :: c : d \qquad e : f :: g : h}{a + e : b + f :: c + g : d + h}.$$

Notice that we can use s-polymorphisms to compose and decompose proportions:

Example 26. Since S is an s-polymorphism of $(\mathbb{N}, S)$ as has been observed in Example 25, we immediately obtain the following characterization of the analogical proportion relation in $(\mathbb{N}, ::)$:

$$a : b :: c : d \quad \Leftrightarrow \quad a - \min(a, b, c, d) : b - \min(a, b, c, d) :: c - \min(a, b, c, d) : d - \min(a, b, c, d).$$

For example, we can use the fact that S is an s-polymorphism in the following way:

$$\begin{aligned} 2 : 3 :: 5 : 7 \quad &\Leftrightarrow \quad SS0 : SSS0 :: SSSSS0 : SSSSSSS0 \\ &\Leftrightarrow \quad S0 : SS0 :: SSSS0 : SSSSSS0 \\ &\Leftrightarrow \quad 0 : S0 :: SSS0 : SSSSS0 \\ &\Leftrightarrow \quad 0 : 1 :: 3 : 5, \end{aligned}$$

which fails in $(\mathbb{N}, ::)$ since $0 - 1 \neq 3 - 5$. That is, we can decompose each analogical proportion $a : b :: c : d$ in $(\mathbb{N}, ::)$ in such a way that at least one of a, b, c, d is equal to 0.

7. Proportional analogies

In §4, we defined proportional homomorphisms as mappings satisfying the strong proportion-preserving property (cf. Remark 11). In this section, we are interested in a related but different notion of proportion-preserving mapping defined as follows. First, we need the following construction (compare to Definition 2):

Definition 27. Given two proportoids $\mathfrak{P} = (P, ::_{\mathfrak{P}})$ and $\mathfrak{R} = (R, ::_{\mathfrak{R}})$, we construct a ***pair proportoid*** (or ***pproportoid***) $\mathfrak{PR} = (P, R, ::_{\mathfrak{PR}})$, where $::_{\mathfrak{PR}} \subseteq P^2 \times R^2$ is a 4-ary analogical proportion relation on P and R satisfying, for all $a, b \in P \cap R$:

$$a : b ::_{\mathfrak{PR}} a : b \quad \text{(reflexivity; r)}, \tag{16}$$

and for all $a, b \in P$ and $c, d \in R$:

$$a : b ::_{\mathfrak{PR}} c : d \quad \Leftrightarrow \quad c : d ::_{\mathfrak{RP}} a : b \quad \text{(symmetry; s)}, \tag{17}$$

$$a : b ::_{\mathfrak{PR}} c : d \quad \Leftrightarrow \quad b : a ::_{\mathfrak{PR}} d : c \quad \text{(inner symmetry; y)}. \tag{18}$$

Notice that every proportoid $(P, ::_{\mathfrak{P}})$ can be turned into a pproportoid $(P, P, ::_{\mathfrak{PP}})$ and we will not distinguish between the two.

Moreover, we consider the following properties for all elements from the appropriate sets:

$$a : a ::_{\mathfrak{PR}} c : c \quad \text{(inner reflexivity; e)},$$

$$a : b ::_{\mathfrak{PR}} c : d \quad \Leftrightarrow \quad a : c ::_{\mathfrak{PR}} b : d \quad \text{(central permutation; c)},$$

$$a : a ::_{\mathfrak{PR}} a : d \quad \Leftrightarrow \quad d = a \quad \text{(determinism; d)},$$

$$a : b ::_{\mathfrak{PR}} c : d \quad \text{and} \quad b : e ::_{\mathfrak{PR}} d : f \quad \Rightarrow \quad a : e ::_{\mathfrak{PR}} c : f \quad \text{(inner transitivity; i)}.$$

Finally, let $\mathfrak{Q} = (Q, ::_{\mathfrak{Q}})$ be an additional proportoid:

$$a : b ::_{\mathfrak{PR}} c : d \quad \text{and} \quad c : d ::_{\mathfrak{RQ}} e : f \quad \Rightarrow \quad a : b ::_{\mathfrak{PQ}} e : f \quad \text{(transitivity; t)}. \tag{19}$$

As for proportoids, we add additional symbols to denote pproportoids satisfying specific properties; for example, an ***i-pproportoid*** is a pproportoid satisfying inner transitivity.

A ***ppt-triple*** is a triple of pproportoids

$$\mathfrak{PRQ} = (P, R, ::_{\mathfrak{PR}})(R, Q, ::_{\mathfrak{RQ}})(P, Q, ::_{\mathfrak{PQ}})$$

satisfying transitivity (19).

Finally, we define

$$\mathcal{S}_{\mathfrak{PR}}(a : b :: c : x) := \{d \in R \mid a : b ::_{\mathfrak{PR}} c : d\}.$$

We will often omit the indices from notation.

Definition 28. A ***(proportional) analogy*** between the proportoids $\mathfrak{P} = (P, ::_{\mathfrak{P}})$ and $\mathfrak{R} = (R, ::_{\mathfrak{R}})$ is any mapping $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ satisfying in the pproportoid $\mathfrak{PR} = (P, R, ::_{\mathfrak{PR}})$ the properties

$$a : b ::_{\mathfrak{PR}} \mathsf{A}a : \mathsf{A}b, \quad \text{for all } a, b \in P.$$

We say that $\mathfrak{P}$ and $\mathfrak{R}$ are ***(proportionally) analogous*** — in symbols, $\mathfrak{P} :: \mathfrak{R}$ — iff there is a bijective analogy from $\mathfrak{P}$ to $\mathfrak{R}$.

Fact 29. *The identity function is an analogy in any e-proportoid satisfying inner reflexivity* (4).

Example 30. The k-successor function S^k is an analogy in $(\mathbb{N}, ::)$ defined as in Example 6.

Example 31. The negation operation $\neg$ is an analogy on $(\{0, 1\}, ::)$ defined as in Example 14.

Example 32. The analogies in $(P, ::_{\leq})$ as defined in Example 4 are the monotone functions on P.

Example 33. The analogies on proportoids arising from metric spaces as in Example 7 are distance-preserving mappings (i.e. isometries).

7.1. **Strong proportion-preserving property.** Recall that homomorphisms satisfy the strong proportion-preserving property (cf. Remark 11)

$$a : b ::_{\mathfrak{P}} c : d \quad \Leftrightarrow \quad \mathsf{H}a : \mathsf{H}b ::_{\mathfrak{R}} \mathsf{H}c : \mathsf{H}d$$

for all $a, b, c, d \in P$. We have the following important result relating homomorphisms and analogies:

Theorem 34. *Every analogy on a ppt-triple* $\mathfrak{PRP}$ *satisfies the strong proportion-preserving property.*

Proof. Let $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ be an analogy. We only prove the direction from left to right with the other direction being analogous: for any $a, b, c, d \in P$, we have

$$\dfrac{\dfrac{\dfrac{a:b ::_{\mathfrak{P}} c:d \qquad c:d ::_{\mathfrak{PR}} \mathsf{A}c:\mathsf{A}d}{a:b ::_{\mathfrak{PR}} \mathsf{A}c:\mathsf{A}d}\,\mathrm{t}}{\mathsf{A}c:\mathsf{A}d ::_{\mathfrak{RP}} a:b}\,\mathrm{s} \qquad a:b ::_{\mathfrak{PR}} \mathsf{A}a:\mathsf{A}b}{\mathsf{A}c:\mathsf{A}d ::_{\mathfrak{R}} \mathsf{A}a:\mathsf{A}b}\,\mathrm{t}}{\mathsf{A}a:\mathsf{A}b ::_{\mathfrak{R}} \mathsf{A}c:\mathsf{A}d.}\,\mathrm{s}$$

□

7.2. **Second injectivity theorem.** Interestingly, the next result shows that analogies are injective in the wide range of dt-proportoids as an analogue to the First Injectivity Theorem 16:

Theorem 35 (Second Injectivity Theorem)**.** *Every analogy on a ppt-triple $\mathfrak{PRP}$, with $\mathfrak{P}$ a d-proportoid, is injective.*

Proof. Let $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ be an analogy. We show that $\mathsf{A}a = \mathsf{A}b$ implies $a = b$, for any $a, b \in P$, by the following derivation:

$$\dfrac{\dfrac{\dfrac{\dfrac{a:b :: \mathsf{A}a:\mathsf{A}b}{a:b :: \mathsf{A}a:\mathsf{A}a}\,\mathsf{A}a=\mathsf{A}b \qquad \mathsf{A}a:\mathsf{A}a :: a:a}{a:b :: a:a}\,\mathrm{t}}{a:a :: a:b}\,\mathrm{s}}{a=b.}\,\mathrm{d}$$

□

7.3. **Closedness under composition.** The composition of two analogies yields another analogy given that the underlying proportoid is transitive which is shown in the next result:

Theorem 36. *The space of all analogies on a t-proportoid forms a monoid with respect to function composition with the neutral element given by the identity function.*

Proof. First, it follows from the definition that analogies are closed under composition in case the underlying algebra is transitive, that is, if $\mathsf{A} : \mathfrak{P} \to \mathfrak{P}$ and $\mathsf{B} : \mathfrak{P} \to \mathfrak{P}$ are analogies, then $\mathsf{B} \circ \mathsf{A}$ is an analogy as well by the following derivation:

$$\dfrac{a:b :: \mathsf{A}a:\mathsf{A}b \qquad \mathsf{A}a:\mathsf{A}b :: \mathsf{BA}a:\mathsf{BA}b}{a:b :: \mathsf{BA}a:\mathsf{BA}b}\,\mathrm{t}.$$

The identity function is an analogy as an immediate consequence of reflexivity. □

7.4. **Proportional idempotency.** Every unary function on a proportoid can be applied repeatedly, which motivates the following definition:

Definition 37. We say that $\mathsf{F} : \mathfrak{P} \to \mathfrak{P}$ is ***(proportionally) idempotent*** iff

$$\mathsf{F}a:\mathsf{F}b :: \mathsf{FF}a:\mathsf{FF}b, \quad \text{holds for all } a, b \in P.$$

Fact 38. *Every analogy is idempotent.*

Remark 39. Notice that by symmetry (2), every idempotent function $\mathsf{F} : \mathfrak{P} \to \mathfrak{P}$ on a t-proportoid $\mathfrak{P}$ satisfies

$$\mathsf{F}^m a : \mathsf{F}^m b :: \mathsf{F}^n a : \mathsf{F}^n b \quad \text{for all } m, n \geq 0 \text{ and } a, b \in P.$$

8. Partial proportional analogies

In this section, we provide a procedure for constructing *partial* proportional analogies from a given enumeration of the source proportoid and selection functions on subsets of the target proportoid (which exist by the presumed axiom of choice).

Concretely, let $\mathfrak{PR}$ be a denumerable pair of i-proportoids satisfying inner transitivity (9), let $\sigma : 2^R \to R \cup \{\mathbf{u}\}$ be a selection function (recall that $\mathbf{u}$ stands for "undefined") such that for every $S \subseteq R$,

$$\sigma S = \mathbf{u} \quad :\Leftrightarrow \quad S = \emptyset, \tag{20}$$

and let e be an enumeration of P. We define

$$\mathcal{S}_{\mathfrak{PR}}(a : b :: \mathbf{u} : x) := \emptyset,$$

for all $a, b \in P$.

We are now ready to introduce the main notion of this section:

Definition 40. Define the partial analogy $\mathsf{A}_{\sigma,e} : \mathfrak{P} \to \mathfrak{R}$, for a selection function σ and enumeration e, inductively by

$$\begin{aligned} \mathsf{A}_{\sigma,e} e_1 &:= \sigma S, \\ \mathsf{A}_{\sigma,e} e_{i+1} &:= \sigma \mathcal{S}_{\mathfrak{PR}}(e_i : e_{i+1} :: \mathsf{A}_{\sigma,e} e_i : x), \quad i \geq 1. \end{aligned}$$

Notice that the domain of $\mathsf{A}_{\sigma,e}$ cannot be empty since we always have $R \neq \emptyset$ by assumption and thus

$$e_1 \in dom\, \mathsf{A}_{\sigma,e}.$$

The next result shows that the construction of Definition 40 always yields a partial analogy:

Theorem 41. *On any i-pproportoid $\mathfrak{PR}$, the partial function $\mathsf{A}_{\sigma,e} : \mathfrak{P} \to \mathfrak{R}$ is an analogy on its non-empty domain, for every choice of e and σ.*

Proof. We need to show

$$a : b ::_{\mathfrak{PR}} \mathsf{A}_{\sigma,e}a : \mathsf{A}_{\sigma,e}b, \quad \text{for all } a, b \in dom\, \mathsf{A}_{\sigma,e}. \tag{21}$$

By definition, we have

$$a = e_{i_a} \quad \text{and} \quad b = e_{i_b}, \quad \text{for some } i_a, i_b \in \mathbb{N}.$$

Without loss of generality, we can assume $i_a \leq i_b$ since otherwise we can apply inner symmetry to obtain $b : a :: \mathsf{A}_{\sigma,e}b : \mathsf{A}_{\sigma,e}a$. So we have

$$i_b = i_a + j, \quad \text{for some } j \in \mathbb{N}.$$

By definition of $\mathsf{A}_{e,\sigma}$, we have

$$\begin{gathered} e_{i_a} : e_{i_a+1} ::_{\mathfrak{PR}} \mathsf{A}_{\sigma,e}e_{i_a} : \mathsf{A}_{\sigma,e}e_{i_a+1} \\ \vdots \\ e_{i_a+j-1} : e_{i_a+j} ::_{\mathfrak{PR}} \mathsf{A}_{\sigma,e}e_{i_a+j-1} : \mathsf{A}_{\sigma,e}e_{i_a+j} \end{gathered}$$

and

$$e_{i_a+k} \in dom\, \mathsf{A}_{\sigma,e}, \quad \text{for all } 1 \leq k \leq j.$$

By the assumed inner transitivity (9) axiom, we obtain

$$e_{i_a} : e_{i_a+j} ::_{\mathfrak{PR}} \mathsf{A}_{\sigma,e}e_{i_a} : \mathsf{A}_{\sigma,e}e_{i_a+j}$$

which is equivalent to (21). □

Example 42. Let e be the identity on $\mathbb{N}$, let $\sigma : 2^{\mathbb{N}} \to \mathbb{N} \cup \{\mathbf{u}\}$ be a selection function, and let S be the unary successor function. Recall the difference proportion relation in $(\mathbb{N}, ::)$ of Example 6

$$a : b :: c : d \quad :\Leftrightarrow \quad a - b = c - d.$$

Definition 40 yields (recall that we have chosen e to be the identity function on $\mathbb{N}$ and it is thus omitted)

$$\begin{aligned} \mathsf{A}_\sigma 1 &= \sigma\mathbb{N} \\ \mathsf{A}_\sigma 2 &= \sigma\mathcal{S}_{(\mathbb{N},S)}(1 : 2 :: \mathsf{A}_\sigma 1 : x) \\ &= \sigma\{2 - 1 + \mathsf{A}_\sigma 1\} \\ &= 1 + \mathsf{A}_{\sigma,e}1 \\ \mathsf{A}_\sigma 3 &= \sigma\mathcal{S}_{(\mathbb{N},S)}(2 : 3 :: \mathsf{A}_\sigma 2 : x) \\ &= \sigma\mathcal{S}_{(\mathbb{N},S)}(2 : 3 :: 1 + \mathsf{A}_\sigma 1 : x) \\ &= \sigma\{3 - 2 + 1 + \mathsf{A}_\sigma 1\} \\ &= 2 + \mathsf{A}_\sigma 1 \\ &\vdots \\ \mathsf{A}_\sigma i &= i + \mathsf{A}_\sigma 1, \quad \text{for all } i \in \mathbb{N}. \end{aligned}$$

Since

$$dom\, \mathsf{A}_\sigma = \mathbb{N},$$

the function $\mathsf{A}_\sigma : \mathbb{N} \to \mathbb{N}$ is a total analogy.

Fact 43. *Every partial analogy on a ppt-triple $\mathfrak{PRP}$ satisfies the strong proportion-preserving property on its domain.*

Proof. Analogous to the proof of Theorem 34. □

Fact 43 tells us that our above procedure for constructing partial analogies automatically yields a procedure for constructing functions satisfying the strong proportion-preserving property and thus for constructing analogy-preserving functions as studied by [6] and [7].

Definition 44. We define the ***cardinality*** of a partial analogy $\mathsf{A}_{\sigma,e}$ by

$$\#\mathsf{A}_{\sigma,e} := \begin{cases} \min\{i \in \mathbb{N} \mid \mathsf{A}_{\sigma,e}e_i = \mathbf{u}\} - 1 & \mathsf{A}_{\sigma,e} \text{ is partial,} \\ \infty & \mathsf{A}_{\sigma,e} \text{ is total.} \end{cases}$$

We call $\mathsf{A}_{\sigma,e}$ ***σ-maximal*** iff there is no analogy $\mathsf{A}_{\sigma',e}$ — with respect to the *same* enumeration e — such that

$$dom\,\mathsf{A}_{\sigma,e} \subsetneq dom\,\mathsf{A}_{\sigma',e}.$$

Fact 45. *$\mathsf{A}_{\sigma,e}$ is σ-maximal iff $\#\mathsf{A}_{\sigma,e}$ is maximal with respect to σ. Every total analogy $\mathsf{A}_{\sigma,e}$ is σ-maximal.*

Proof. An immediate consequence of the fact that by construction,

$$\mathsf{A}_{\sigma,e}e_i = \mathbf{u} \quad \text{implies} \quad \mathsf{A}_{\sigma,e}e_j = \mathbf{u}, \quad \text{for all } j \geq i. \tag{22}$$

□

We shall now show that *every* analogy is constructed from an enumeration and a selection function as above.

Theorem 46. *For every partial analogy $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ there is an enumeration e of P and a selection function $\sigma : 2^R \to R \cup \{\boldsymbol{u}\}$ such that $\mathsf{A} = \mathsf{A}_{\sigma,e}$.*

Proof. Let e be an enumeration of P enumerating first the elements of $dom\,\mathsf{A}$ and then the rest in arbitrary order. Define, for every $i \geq 1$,

$$\mathsf{A}_{\sigma,e}e_1 := \sigma R := \mathsf{A}e_1,$$
$$\mathsf{A}_{\sigma,e}e_{i+1} := \begin{cases} \sigma\mathcal{S}_{\mathfrak{P}\mathfrak{R}}(e_i : e_{i+1} :: \mathsf{A}e_i : x) := \mathsf{A}e_{i+1} & e_{i+1} \in dom\,\mathsf{A}, \\ \mathbf{u} & \text{otherwise.} \end{cases}$$

We clearly have $\mathsf{A}_{\sigma,e} = \mathsf{A}$ by construction. □

Notice that σ-maximality is defined with respect to σ for some fixed e — this does *not* gurantee that there is no analogy A constructed by other means with $dom\,\mathsf{A}_{\sigma,e} \subsetneq dom\,\mathsf{A}$. We say that $\mathsf{A}_{\sigma,e}$ is ***maximal*** iff there is no such analogy A.

Corollary 47. *A partial analogy $\mathsf{A}_{\sigma,e}$ is maximal iff there is no analogy $\mathsf{A}_{\sigma',e'}$ such that $\#\mathsf{A}_{\sigma,e} < \#\mathsf{A}_{\sigma',e'}$.*

Proof. A direct consequence of Fact 45 and Theorem 46. □

Let $\mathsf{A}_{\sigma,e}$ be an arbitrary partial analogy and let n be the cardinality of $\mathsf{A}_{\sigma,e}$. Suppose there is some $i \in \mathbb{N}$ such that

$$e_i \notin dom\,\mathsf{A}_{\sigma,e}$$

satisfying

$$\mathcal{S}(e_n : e_i :: \mathsf{A}e_n : x) \neq \emptyset. \tag{23}$$

Define the transposition $\tau : \mathbb{N} \to \mathbb{N}$ by

$$\tau n := i$$
$$\tau i := n$$
$$\tau j := j, \quad \text{for all other } j \in \mathbb{N}.$$

We then have by (23)

$$\mathsf{A}_{\sigma,e_\tau}e_{\tau n} = \sigma\mathcal{S}(e_n : e_i :: \mathsf{A}e_n : x) \neq \mathbf{u},$$

which entails

$$\#\mathsf{A}_{\sigma,e_\tau} = n + 1 = \#\mathsf{A}_{\sigma,e} + 1,$$

and thus

$$dom\,\mathsf{A}_{\sigma,e} \subsetneq dom\,\mathsf{A}_{\sigma,e_\tau}.$$

In other words, we have increased the cardinality of the partial analogy $\mathsf{A}_{\sigma,e}$ by one by transforming it into the partial analogy $\mathsf{A}_{\sigma,e_\tau}$. This procedure can be iterated until a (partial) analogy has been reached whose cardinality can no longer be increased.

9. Proportional identity

In this section, we shall introduce a proportional identity relation (cf. Definition 53). First, we introduce an auxiliary identity relation:

Definition 48. Let $a, b \in P$. We define

$$a =_c b \quad :\Leftrightarrow \quad a : b :: c : c, \quad \text{for some } c \in P.$$

Fact 49. $\dfrac{a = b}{a =_c b}$.

Proposition 50. *In any cet-proportoid, the relation $=_c$ is an equivalence relation for any element c.*

Proof. The reflexivity and symmetry of $=_c$ follow from inner symmetry (3) and the assumed inner reflexivity (4) of the analogical proportion relation, respectively. To prove transitivity, we show the implication

$$\frac{a =_c b \qquad b =_c d}{a =_c d}$$

by the following derivation:

$$\dfrac{\dfrac{\dfrac{a =_c b}{a : b :: c : c}\,\mathrm{c}}{a : c :: b : c} \qquad \dfrac{\dfrac{\dfrac{\dfrac{b =_c d}{b : d :: c : c}\,\mathrm{y}}{d : b :: c : c}\,\mathrm{c}}{d : c :: b : c}\,\mathrm{s}}{b : c :: d : c}\,\mathrm{t}}{\dfrac{\dfrac{a : c :: d : c}{a : d :: c : c}\,\mathrm{c}}{a =_c d.}}$$

□

Proposition 51. *Every n-ary polymorphism f satisfies, for any elements $a_1, b_1, \ldots, a_n, b_n \in P$,*

$$\frac{a_1 =_c b_1 \quad \ldots \quad a_n =_c b_n}{fa_1 \ldots a_n =_{fc\ldots c} fb_1 \ldots b_n.}$$

Proof.

$$\dfrac{\dfrac{\dfrac{a_1 =_c b_1 \quad \ldots \quad a_n =_c b_n}{a_1 : b_1 :: c : c \quad \ldots \quad a_n : b_n :: c : c}}{fa_1 \ldots a_n : fb_1 \ldots b_n :: fc \ldots c : fc \ldots c}}{fa_1 \ldots a_n =_{fc\ldots c} fb_1 \ldots b_n.}$$

□

Proposition 52. *In any ct-proportoid, we have for any elements $a, b, c, d, e, f, g, h, i \in P$,*

$$\frac{a =_i e \quad b =_i f \quad c =_i g \quad d =_i h \qquad a : b :: c : d}{e : f :: g : h}\,ct.$$

Proof.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{c =_i g}{c : g :: i : i} \qquad \dfrac{d =_i h}{d : h :: i : i}\,\mathrm{s}}{i : i :: d : h}\,\mathrm{t}}{c : g :: d : h}\,\mathrm{c}}{c : d :: g : h}\,\mathrm{s}}{g : h :: c : d} \qquad \dfrac{a : b :: c : d}{c : d :: a : b}\,\mathrm{s}}{g : h :: a : b}\,\mathrm{t} \qquad \dfrac{\dfrac{\dfrac{a =_i e}{a : e :: i : i} \qquad \dfrac{b =_i f}{b : f :: i : i}\,\mathrm{s}}{i : i :: b : f}\,\mathrm{t}}{a : e :: b : f}\,\mathrm{c}}{a : b :: e : f}\,\mathrm{t}$$

$$\frac{g : h :: e : f}{e : f :: g : h.}\,\mathrm{s}$$

□

We now generalize Definition 48 as follows:

Definition 53. Let $a, b \in P$. We define the ***proportional identity relation*** (or ***p-identity***) by

$$a \ddot{=} b \quad :\Leftrightarrow \quad a =_c b, \quad \text{for some } c \in P,$$

In case $a \ddot{=} b$, we call a and b ***proportionally identical*** (or ***p-identical***).

The following definition is analogous to the definition of a homomorphism kernel (cf. Definition 20):

Definition 54. We define the ***kernel*** of an analogy $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ by

$$ker\ \mathsf{A} := \left\{ab \in P^2 \;\middle|\; \mathsf{A}a = \mathsf{A}b\right\}.$$

We have the following implications:

Proposition 55. $\dfrac{ab \in ker\ \mathsf{A}}{a \mathrel{\ddot{=}} b}$.

Proof.

$$\dfrac{\dfrac{\mathsf{A} \text{ is an analogy}}{a : b :: \mathsf{A}a : \mathsf{A}b} \qquad \dfrac{ab \in ker\ \mathsf{A}}{\mathsf{A}a = \mathsf{A}b}}{\dfrac{a : b :: \mathsf{A}a : \mathsf{A}a}{a \mathrel{\ddot{=}} b.}}$$

□

Proposition 56. $\dfrac{ab \in ker\ \mathsf{A} \qquad cd \in ker\ \mathsf{A}}{a : b :: c : d}\ t.$

Proof.

$$\dfrac{\dfrac{ab \in ker\ \mathsf{A}}{a \mathrel{\ddot{=}} b}\ 55 \qquad \dfrac{cd \in ker\ \mathsf{A}}{c \mathrel{\ddot{=}} d}\ 55}{a : b :: c : d.}\ \text{t, } 61$$

□

Proposition 57. *In any et-proportoid, we have $a \mathrel{\ddot{=}} b$ iff $a =_c b$ for all c.*

Proof. We only have to prove the direction from left to right. By definition of identity, the assumption $a \mathrel{\ddot{=}} b$ implies that there is some d such that $a : b :: d : d$. By the assumed inner reflexivity (4), we have $d : d :: c : c$. Now apply transitivity to $a : b :: d : d$ and $d : d :: c : c$ to obtain $a : b :: c : c$ which is equivalent to $a =_c b$. □

Proposition 58. *$a = b$ implies $a \mathrel{\ddot{=}} b$ in any proportoid, and $a \mathrel{\ddot{=}} b$ implies $a = b$ in any det-proportoid.*

Proof. The first implication is a direct consequence of inner reflexivity (4) which implies $a : a :: c : c$, for every c. The second implication follows from:

$$\begin{aligned}
a \mathrel{\ddot{=}} b \quad &\Leftrightarrow \quad a : b :: c : c, \quad \text{for some } c, \\
&\overset{et}{\Leftrightarrow} \quad a : b :: c : c, \quad \text{for all } c, \quad \text{(Proposition 57)} \\
&\Leftrightarrow \quad a : b :: a : a \\
&\overset{d}{\Leftrightarrow} \quad a = b.
\end{aligned}$$

□

Lemma 59. $\dfrac{a : b :: e : e \qquad c : d :: f : f}{a : b :: c : d}\ t.$

Proof.

$$\dfrac{\dfrac{a : b :: e : e}{} \qquad \dfrac{\dfrac{e : e :: f : f \qquad \dfrac{c : d :: f : f}{f : f :: c : d}\ \text{s}}{e : e :: c : d}\ \text{t}}{a : b :: c : d.}\ \text{t}$$

□

Theorem 60. *The proportional identity relation is a congruence in any cet-proportoid.*

Proof. That identity is an equivalence relation is an immediate consequence of Propositions 50 and 57 (where we had to assume the "e" (4) in "cet").

It remains to show:

$$\dfrac{a \mathrel{\ddot{=}} e \quad b \mathrel{\ddot{=}} f \quad c \mathrel{\ddot{=}} g \quad d \mathrel{\ddot{=}} h \qquad a : b :: c : d}{e : f :: g : h.}$$

To prove this implication, we proceed similar to the proof of Proposition 52:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{c \ddot{=} g}{c : g :: i : i} \quad \dfrac{d \ddot{=} h}{d : h :: j : j}}{c : g :: d : h}\,59}{c : d :: g : h}\,\mathrm{c}}{g : h :: c : d}\,\mathrm{s} \quad \dfrac{a : b :: c : d}{c : d :: a : b}\,\mathrm{s}}{g : h :: a : b}\,\mathrm{t} \quad \dfrac{\dfrac{\dfrac{a \ddot{=} e}{a : e :: k : k} \quad \dfrac{b \ddot{=} f}{b : f :: l : l}}{a : e :: b : f}\,59}{a : b :: e : f}\,\mathrm{c}}{\dfrac{g : h :: e : f}{e : f :: g : h.}\,\mathrm{s}}\,\mathrm{t}$$

□

Theorem 61. $\dfrac{a \ddot{=} b \qquad c \ddot{=} d}{a : b :: c : d}\ t.$

Proof.

$$\dfrac{\dfrac{a \ddot{=} b}{a : b :: e : e} \quad \dfrac{e : e :: f : f \qquad \dfrac{\dfrac{c \ddot{=} d}{c : d :: f : f}}{f : f :: c : d}\,\mathrm{s}}{e : e :: c : d}\,\mathrm{t}}{a : b :: c : d.}\,\mathrm{t}$$

□

Proposition 62. *Every proportional polymorphism $f : P^n \to P$ satisfies*

$$\dfrac{a_1 \ddot{=} b_1 \quad \dots \quad a_n \ddot{=} b_n}{f a_1 \dots a_n \ddot{=} f b_1 \dots b_n.}$$

Proof.

$$\dfrac{\dfrac{\dfrac{a_1 \ddot{=} b_1 \quad \dots \quad a_n \ddot{=} b_n}{a_1 : b_1 :: c_1 : c_1 \quad \dots \quad a_n : b_n :: c_n : c_n}}{f a_1 \dots a_n : f b_1 \dots b_n :: f c_1 \dots c_n : f c_1 \dots c_n}}{f a_1 \dots a_n \ddot{=} f b_1 \dots b_n.}$$

□

Theorem 63. *For any analogy $\mathsf{A} : \mathfrak{P} \to \mathfrak{P}$ on a cft-proportoid $\mathfrak{P}$ and any injective function $f : P \to P$,*

$$\mathsf{A} f a \ddot{=} f \mathsf{A} a, \quad \textit{for all } a \in P.$$

Proof. Since A is an analogy by assumption, we have the following derivation, for all $a \in P$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{f \text{ is injective}}{\mathsf{A}a : f\mathsf{A}a :: a : fa}\,\mathrm{f} \quad \dfrac{\mathsf{A} \text{ is an analogy}}{a : fa :: \mathsf{A}a : \mathsf{A}fa}}{\mathsf{A}a : f\mathsf{A}a :: \mathsf{A}a : \mathsf{A}fa}\,\mathrm{t}}{\mathsf{A}a : \mathsf{A}a :: \mathsf{A}fa : f\mathsf{A}a}\,\mathrm{c}}{\mathsf{A}fa : f\mathsf{A}a :: \mathsf{A}a : \mathsf{A}a}\,\mathrm{s}}{\mathsf{A}fa =_{\mathsf{A}a} f\mathsf{A}a}}{\mathsf{A}fa \ddot{=} f\mathsf{A}a.}$$

□

Proposition 64. $\dfrac{a \ddot{=} b}{\mathsf{P}a \ddot{=} \mathsf{P}b}$ *holds for any pp-mapping* P.

Proof.

$$\dfrac{\dfrac{\dfrac{a \ddot{=} b}{a : b :: c : c \quad \text{for some } c}}{\mathsf{P}a : \mathsf{P}b :: \mathsf{P}c : \mathsf{P}c \quad \text{for some } c}}{\mathsf{P}a \ddot{=} \mathsf{P}b.}\ (15)$$

□

Proposition 65. *For any homomorphism $\mathsf{H} : \mathfrak{P} \to \mathfrak{R}$,*

$$a \ddot{=} b \quad \Leftrightarrow \quad \mathsf{H}a \ddot{=} \mathsf{H}b.$$

Proof. We have

$$\begin{aligned} a \mathrel{\ddot{=}} b \quad &\Leftrightarrow \quad a : b :: c : c \quad \text{for some } c \\ &\Leftrightarrow \quad \mathsf{H}a : \mathsf{H}b :: \mathsf{H}c : \mathsf{H}c \quad \text{for some } c \\ &\Leftrightarrow \quad \mathsf{H}a \mathrel{\ddot{=}} \mathsf{H}b. \end{aligned}$$

□

Definition 66. We call $a \in P$ a ***p-fixed point*** of $\mathsf{F} : P \to P$ iff $a \mathrel{\ddot{=}} \mathsf{F}a$.

Proposition 67. *Let* A *be an analogy with a fixed point on a c-proportoid* $(P, ::)$*. Then every element of* P *is a p-fixed point of* A*.*

Proof. Let a be an arbitrary element of P, and let $b \in P$ be a fixed point of A, that is, $\mathsf{A}b = b$. We then have

$$\dfrac{\dfrac{\dfrac{\dfrac{\mathsf{A} \text{ is an analogy}}{a : b :: \mathsf{A}a : \mathsf{A}b}\,\mathrm{c}}{a : \mathsf{A}a :: b : \mathsf{A}b}\,\mathsf{A}b = b}{a : \mathsf{A}a :: b : b}}{a \mathrel{\ddot{=}} Aa.}$$

□

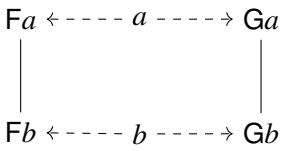
10. Proportional function relations

We now turn our attention to functions on proportoids where we wish to be able to compare two functions with respect to the analogical proportion relation.

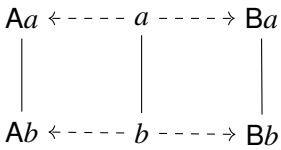
10.1. **Functional proportionality.** Given $\mathsf{F} : \mathfrak{P} \to \mathfrak{R}$ and $\mathsf{G} : \mathfrak{P} \to \mathfrak{Q}$, define their ***functional proportionality relation*** by

$$\mathsf{F} :: \mathsf{G} \quad :\Leftrightarrow \quad \mathsf{F}a : \mathsf{F}b ::_{\mathfrak{RQ}} \mathsf{G}a : \mathsf{G}b, \quad \text{for all } a, b \in P.$$

In case $\mathsf{F} :: \mathsf{G}$, we say that F and G are ***proportional***. This can be depicted as follows:



In case A and B are analogies, the figure for $\mathsf{A} :: \mathsf{B}$ can be refined to:



Fact 68. $\mathsf{F} :: \mathsf{I}$ *iff* F *is an analogy.*

Before we show that functional proportionality is a congruence relation, we shall first prove an auxiliary lemma which is interesting in its own right as it shows that functional proportionality is in a sense compatible with analogical proportions.

Lemma 69. *For any analogies* $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ *and* $\mathsf{B} : \mathfrak{P} \to \mathfrak{Q}$ *on a ppt-triple* $\mathfrak{QQR}$*,*

$$\dfrac{a : b ::_{\mathfrak{P}} c : d \qquad \mathsf{A}a : \mathsf{A}b ::_{\mathfrak{RQ}} \mathsf{B}a : \mathsf{B}b}{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{RQ}} \mathsf{B}c : \mathsf{B}d}\,t.$$

Proof.

$$\dfrac{\dfrac{\dfrac{a : b ::_{\mathfrak{P}} c : d}{\mathsf{B}a : \mathsf{B}b ::_{\mathfrak{Q}} \mathsf{B}c : \mathsf{B}d}\,\mathrm{s}}{\mathsf{B}c : \mathsf{B}d ::_{\mathfrak{Q}} \mathsf{B}a : \mathsf{B}b}\,\mathrm{s}\ \text{PPP 34} \qquad \dfrac{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{RQ}} \mathsf{B}a : \mathsf{B}b}{\mathsf{B}a : \mathsf{B}b ::_{\mathfrak{QR}} \mathsf{A}a : \mathsf{A}b}\,\mathrm{s}}{\dfrac{\mathsf{B}c : \mathsf{B}d ::_{\mathfrak{QR}} \mathsf{A}a : \mathsf{A}b}{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{RQ}} \mathsf{B}c : \mathsf{B}d.}\,\mathrm{s}}\,\mathrm{t}$$

□

Theorem 70. *Functional proportionality of analogies is a congruence on any t-proportoid.*

Proof. Reflexivity, symmetry, and transitivity of the proportionality relation follows by reflexivity, symmetry, and the assumed transitivity of analogical proportions. It remains to show that it is compatible with composition:

$$\dfrac{\dfrac{\dfrac{\mathsf{A} :: \mathsf{B}}{\mathsf{A}a : \mathsf{A}b :: \mathsf{B}a : \mathsf{B}b} \quad \dfrac{\mathsf{C} :: \mathsf{D}}{\mathsf{C}a : \mathsf{C}b :: \mathsf{D}a : \mathsf{D}b}}{\mathsf{AC}a : \mathsf{AC}b :: \mathsf{BD}a : \mathsf{BD}b}}{\mathsf{AC} :: \mathsf{BD}.}\ \text{t, 69}$$

□

Example 71. All generalized successor functions S^k and S^ℓ are functionally proportional in $(\mathbb{N}, ::)$ (cf. Example 6), that is, $S^k :: S^\ell$ holds for all $k, \ell \geq 0$.

Proposition 72. *Functional proportionality is reflexive and symmetric on any proportoid. If the underlying proportoid is transitive, then functional proportionality is transitive and therefore an equivalence relation.*

Proof. Reflexivity and symmetry follow from the reflexivity and symmetry of analogical proportions, and the assumed transitivity induces transitivity. □

Theorem 73. *Let $\mathfrak{PRR}$ be a ppt-triple, let $\mathsf{A} : \mathfrak{P} \to \mathfrak{R}$ be an analogy, and let $\mathsf{F} : \mathfrak{P} \to \mathfrak{R}$ be an arbitrary function. If $\mathsf{A} :: \mathsf{F}$, then F is an analogy.*

Proof.

$$\dfrac{\dfrac{\mathsf{A}\text{ is an analogy}}{a : b ::_{\mathfrak{PR}} \mathsf{A}a : \mathsf{A}b} \quad \dfrac{\mathsf{A} :: \mathsf{F}}{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{R}} \mathsf{F}a : \mathsf{F}b}}{a : b ::_{\mathfrak{PR}} \mathsf{F}a : \mathsf{F}b.}\ \text{t}$$

□

Theorem 74. *All analogies $\mathsf{A}, \mathsf{B} : \mathfrak{P} \to \mathfrak{R}$ on ppt-triples $\mathfrak{RPR}$ are functionally proportional.*

Proof. We have the following derivation, for any $a, b \in P$:

$$\dfrac{\dfrac{\dfrac{a : b ::_{\mathfrak{PR}} \mathsf{A}a : \mathsf{A}b}{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{RP}} a : b}\ \text{s} \quad a : b ::_{\mathfrak{PR}} \mathsf{B}a : \mathsf{B}b}{\mathsf{A}a : \mathsf{A}b ::_{\mathfrak{R}} \mathsf{B}a : \mathsf{B}b}\ \text{t}}{\mathsf{A} :: \mathsf{B}.}$$

□

The next result connects analogies with homomorphisms with respect to functional proportionality:

Theorem 75. *For any analogy A and homomorphism H on a t-proportoid, we have*

$$\mathsf{HA} :: \mathsf{AH}.$$

Proof.

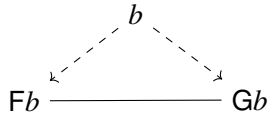
$$\dfrac{\dfrac{\dfrac{\dfrac{\mathsf{A}\text{ is an analogy}}{a : b :: \mathsf{A}a : \mathsf{A}b}}{\mathsf{H}a : \mathsf{H}b :: \mathsf{HA}a : \mathsf{HA}b}\ \mathsf{H}\text{ is a homomorphism}}{\mathsf{HA}a : \mathsf{HA}b :: \mathsf{H}a : \mathsf{H}b}\ \text{s} \quad \dfrac{\mathsf{A}\text{ is an analogy}}{\mathsf{H}a : \mathsf{H}b :: \mathsf{AH} : \mathsf{AH}b}\ \text{t}}{\mathsf{HA}a : \mathsf{HA}b :: \mathsf{AH}a : \mathsf{AH}b}}{\mathsf{HA} :: \mathsf{AH}}$$

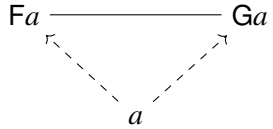
□

10.2. **Diamond equivalence.** Given $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{R}$, define their (***proportional***) ***diamond equivalence*** by

$$\mathsf{F} \cdot :: \cdot \mathsf{G} \quad :\Leftrightarrow \quad \mathsf{F}a : \mathsf{G}a ::_{\mathfrak{R}} \mathsf{F}b : \mathsf{G}b, \quad \text{for all } a, b \in P.$$

This can be depicted as follows:





Notice the similarity to functional proportionality in §10.1.

Proposition 76. *All analogies are diamond equivalent on i-proportoids.*

Proof.

$$\dfrac{\dfrac{\dfrac{\mathsf{A}\text{ is analogy}}{a : \mathsf{A}a :: b : \mathsf{A}b}}{\mathsf{A}a : a :: \mathsf{A}b : b}\,\mathrm{y} \qquad \dfrac{\mathsf{B}\text{ is analogy}}{a : \mathsf{B}a :: b : \mathsf{B}b}}{\dfrac{\mathsf{A}a : \mathsf{B}a :: \mathsf{A}b : \mathsf{B}b}{\mathsf{A} \cdot : \cdot \mathsf{B}.}}\,\mathrm{i}$$

□

Fact 77. $\mathsf{A} \cdot : \cdot \mathsf{I}$ *iff* A *is an analogy.*

Proposition 78. *Diamond equivalence is an equivalence relation on i-proportoids.*

Proof. Reflexivity of diamond follows from inner reflexivity of analogical proportions, and symmetry holds trivially. The following derivation proves transitivity of diamond:

$$\dfrac{\dfrac{\mathsf{F} \cdot : \cdot G}{\mathsf{F}a : \mathsf{G}a :: \mathsf{F}b : \mathsf{G}b} \qquad \dfrac{G \cdot : \cdot \mathsf{H}}{\mathsf{G}a : \mathsf{H}a :: \mathsf{G}b : \mathsf{H}b}}{\dfrac{\mathsf{F}a : \mathsf{H}a :: \mathsf{F}b : \mathsf{H}b}{\mathsf{F} \cdot : \cdot \mathsf{H}.}}\,\mathrm{i}$$

□

Fact 79. *In any c-proportoid, we have* $\mathsf{F} \cdot : \cdot G$ *iff* $\mathsf{F} :: \mathsf{G}$.

Fact 80. *Diamond equivalence is a congruence on ct-proportoids.*

Proof. A direct consequence of Theorem 70 and Fact 79. □

Example 81. In $(\mathbb{N}, ::)$ defined as in Example 6, we have

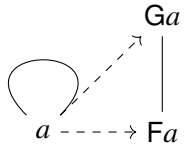
$$\mathsf{F} \cdot : \cdot \mathsf{G} \quad \Leftrightarrow \quad \mathsf{F}a - \mathsf{F}b = \mathsf{G}a - \mathsf{G}b, \quad \text{for all } a, b \in \mathbb{N}.$$

In particular, we have $S^k \cdot : \cdot S^\ell$, for all $k, \ell \geq 0$.

10.3. **Equivalence.** Given mappings $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{R}$, define their (***proportional***) ***equivalence*** by

$$\mathsf{F} \equiv \mathsf{G} \quad :\Leftrightarrow \quad a : a ::_{\mathfrak{P}\mathfrak{R}} \mathsf{F}a : \mathsf{G}a, \quad \text{for all } a \in P.$$

This situation can be depicted as follows:



The following result provides a simple way to show that two mappings are *not* equivalent given that determinism (6) holds:

Theorem 82. *Proportionally equivalent mappings have the same fixed points in d-proportoids.*

Proof. Let a be a fixed point of F thus satisfying $\mathsf{F}a = a$. Then, by determinism (6) and since F and G are equivalent by assumption, we have

$$a : a :: \mathsf{F}a : \mathsf{G}a \quad \Leftrightarrow \quad \mathsf{G}a = a,$$

which means that a is a fixed point of G as well. By inner reflexivity, an analogous argument shows that every fixed point of G is a fixed of F. □

Corollary 83. $\mathsf{F} \equiv \mathsf{I}$ *iff* $\mathsf{F} = \mathsf{I}$ *holds in d-proportoids.*

Proof. A direct consequence of Theorem 82. □

Theorem 84. *Proportional equivalence is an equivalence relation on it-proportoids, and it is a congruence for all analogies on t-proportoids.*

Proof. Reflexivity of proportional equivalence follows from (1), symmetry follows from (3), and transitivity follows from (9).

It remains to show that equivalence is compatible with composition of analogies. In case A is an analogy, we have

$$\dfrac{\dfrac{\dfrac{\mathsf{B} \equiv \mathsf{C}}{a : a :: \mathsf{B}a : \mathsf{C}a}}{a : a :: \mathsf{AB}a : \mathsf{AC}a}\,\mathsf{A}\text{ is an analogy}}{\mathsf{AB} \equiv \mathsf{AC}.}$$

Moreover, we always have (even if A is not an analogy)

$$\dfrac{\dfrac{\dfrac{\text{inner reflexivity}}{a : a :: \mathsf{A}a : \mathsf{A}a} \quad \dfrac{\mathsf{B} \equiv \mathsf{C}}{\mathsf{A}a : \mathsf{A}a :: \mathsf{BA}a : \mathsf{CA}a}\,\mathrm{t}}{a : a :: \mathsf{BA}a : \mathsf{CA}a}}{\mathsf{BA} \equiv \mathsf{CA}.}$$

This shows that equivalence is left and right compatible, which by Proposition 1 means that it is a congruence. □

Theorem 85. *Proportional equivalence is left cancellative for analogies on any t-proportoid in the strong sense that for all analogies* $\mathsf{A}, \mathsf{B}, \mathsf{C}$*,*

$$\mathsf{AB} \equiv \mathsf{AC} \quad \Leftrightarrow \quad \mathsf{B} \equiv \mathsf{C}.$$

In case A *commutes with* B *and* C *in the sense that*

$$\mathsf{BA}a : \mathsf{CA}a :: \mathsf{AB}a : \mathsf{AC}a, \quad \textit{for all } a \in P, \tag{24}$$

it follows that equivalence is right cancellative in the strong sense as well, that is,

$$\mathsf{BA} \equiv \mathsf{CA} \quad \Leftrightarrow \quad \mathsf{A} \equiv \mathsf{C}.$$

Proof. For the direction from left to right, we have the following derivation:

$$\dfrac{\dfrac{\dfrac{\mathsf{AB} \equiv \mathsf{AC}}{a : a :: \mathsf{AB}a : \mathsf{AC}a} \quad \dfrac{\mathsf{A} \text{ is an analogy}}{\mathsf{AB}a : \mathsf{AC}a :: \mathsf{B}a : \mathsf{C}a}\,\mathrm{t}}{a : a :: \mathsf{B}a : \mathsf{C}a}}{\mathsf{B} \equiv \mathsf{C}.}$$

The direction from right to left holds trivially:

$$\dfrac{\dfrac{\dfrac{\mathsf{B} \equiv \mathsf{C}}{a : a :: \mathsf{B}a : \mathsf{C}a}}{a : a :: \mathsf{AB}a : \mathsf{AC}a}\,\mathsf{A} \text{ is an analogy}}{\mathsf{AB} \equiv \mathsf{AC}.}$$

For the second part, we assume that A commutes with B and C in the sense of (24). The direction from left to right is shown by the following derivation:

$$\dfrac{\dfrac{\dfrac{\dfrac{\mathsf{BA} \equiv \mathsf{CA}}{a : a :: \mathsf{BA}a : \mathsf{CA}a} \quad \dfrac{\mathsf{A} \text{ commutes with } \mathsf{B} \text{ and } \mathsf{C}}{\mathsf{BA}a : \mathsf{CA}a :: \mathsf{AB}a : \mathsf{AC}a}}{a : a :: \mathsf{AB}a : \mathsf{AC}a} \quad \dfrac{\mathsf{A} \text{ is an analogy}}{\mathsf{AB}a : \mathsf{AC}a :: \mathsf{B}a : \mathsf{C}a}\,\mathrm{t}}{a : a : \mathsf{B}a : \mathsf{C}a}}{\mathsf{B} \equiv \mathsf{C}.}$$

For the other direction, we compute:

$$\dfrac{\dfrac{\dfrac{\dfrac{\mathsf{B} \equiv \mathsf{C}}{a : a :: \mathsf{B}a : \mathsf{C}a}}{a : a :: \mathsf{AB}a : \mathsf{AC}a}\,\mathsf{A} \text{ is an analogy} \quad \dfrac{\mathsf{A} \text{ commutes with } \mathsf{B} \text{ and } \mathsf{C}}{\mathsf{AB}a : \mathsf{AC}a :: \mathsf{BA}a : \mathsf{CA}a}\,\mathrm{t}}{a : a :: \mathsf{BA}a : \mathsf{CA}a}}{\mathsf{BA}a \equiv \mathsf{CA}a.}$$

□

Theorem 86. $\dfrac{\mathsf{F} \equiv \mathsf{G}}{\mathsf{F} :: \mathsf{G}}$ *et.*

Proof.

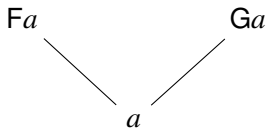
$$\dfrac{\dfrac{\dfrac{\mathsf{F} \equiv \mathsf{G}}{\mathsf{F}a : \mathsf{G}a :: a : a}\,\mathrm{s} \quad \dfrac{\dfrac{\text{inner reflexivity (4)}}{a : a :: b : b}\,\mathrm{e} \quad \dfrac{\mathsf{F} \equiv \mathsf{G}}{b : b :: \mathsf{F}b : \mathsf{G}b}}{a : a :: \mathsf{F}b : \mathsf{G}b}\,\mathrm{t}}{\mathsf{F}a : \mathsf{F}b :: \mathsf{G}a : \mathsf{G}b}}{\mathsf{F} :: \mathsf{G}.}$$

□

10.4. **Join equivalence.** Given $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{P}$, define their ***(proportional) join equivalence*** by

$$\mathsf{F} \vee \mathsf{G} \quad :\Leftrightarrow \quad a : \mathsf{F}a :: a : \mathsf{G}a, \quad \text{for all } a \in P.$$

This can be depicted as follows:



Fact 87. $\mathsf{F} \vee \mathsf{G}$ *iff* $\mathsf{F} \equiv \mathsf{G}$ *in c-proportoids.*

Corollary 88. *Join equivalence is a congruence for analogies on any ct-proportoid.*

Proof. A direct consequence of Theorem 84 and Fact 87. □

Corollary 88 has rather strong assumptions as central permutation often fails in practice and it holds only for analogies on any ct-proportoid. The next result states that for join equivalence to be an equivalence relation (not a congruence), only transitivity is required:

Fact 89. *Join equivalence is an equivalence relation on any t-proportoid.*

The following result gives us a simple method to show that two mappings are *not* join equivalent in case determinism (6) holds:[4]

Proposition 90. *Join equivalent mappings have the same fixed points in d-proportoids.*

Proof. Given join equivalent mappings F, G, determinism implies that for any $a \in P$, in case a is a fixed point of F thus satisfying $\mathsf{F}a = a$,

$$a : \mathsf{F}a :: a : \mathsf{G}a \quad \Leftrightarrow \quad \mathsf{G}a = a,$$

which means that a is a fixed point of G as well. An analogous argument shows that every fixed point of G is a fixed point of F. □

Corollary 91. $\mathsf{F} \vee \mathsf{I}$ *iff* $\mathsf{F} = \mathsf{I}$ *in d-proportoids.*

Proof. A direct consequence of Proposition 90. □

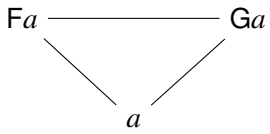
The next observation shows that the converse of Proposition 90 fails in general since $S^k$and S^ℓ have the same fixed points (none) for all $k, \ell \geq 1$:

Example 92. In $(\mathbb{N}, ::)$ defined as in Example 6, we have $S^k \vee S^\ell$ iff $k = \ell$.

10.5. **Triangular relation.** Given $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{P}$, define the (***proportional***) ***triangular relation*** by

$$\mathsf{F} \triangledown \mathsf{G} \quad :\Leftrightarrow \quad \mathsf{F} \vee \mathsf{G} \quad \text{and} \quad a : \mathsf{F}a :: \mathsf{F}a : \mathsf{G}a \quad \text{and} \quad a : \mathsf{G}a :: \mathsf{G}a : \mathsf{F}a, \quad \text{for all } a \in P.$$

This can be depicted as follows (see the similarity to join equivalence in §10.4):



The following observation shows that the triangular relation is in general *not* reflexive:

Proposition 93. $\mathsf{F} \triangledown \mathsf{F}$ *iff* $\mathsf{F} = \mathsf{I}$ *in d-proportoids.*

Proof. We have $\mathsf{F} \triangledown \mathsf{F}$ only if $a : \mathsf{F}a :: \mathsf{F}a : \mathsf{F}a$, for all $a \in P$. By determinism, this is equivalent to $\mathsf{F}a = a$, that is, a is a fixed point of F for *each* $a \in P$, which is equivalent to $\mathsf{F} = \mathsf{I}$. The other direction holds trivially. □

Proposition 94. $\dfrac{\mathsf{F} \triangledown \mathsf{G}}{\mathsf{F} \equiv \mathsf{G}}\ c.$

Proof. $\mathsf{F} \triangledown \mathsf{G}$ implies $\mathsf{F} \vee \mathsf{G}$ which implies $\mathsf{F} \equiv \mathsf{G}$ in all c-proportoids by Fact 87. □

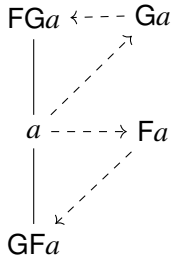
Example 95. As a direct consequence of Example 92, in $(\mathbb{N}, ::)$ we have $S^k \triangledown S^\ell$ iff $k = \ell$.

[4]This is analogous to Theorem 82.

10.6. **Bowtie relation.** Given mappings $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{P}$, define the (***proportional***) ***bowtie relation*** by

$$\mathsf{F} \bowtie \mathsf{G} \quad :\Leftrightarrow \quad a : \mathsf{GF}a :: a : \mathsf{FG}a, \quad \text{for all } a \in P,$$

which can be depicted as follows:



Notice that the bowtie relation is connected to the commutation of F and G and the next result shows that bowtie equivalent mappings commute with respect to proportional equivalence in c-proportoids satisfying central permutation:

Proposition 96. $\mathsf{F} \bowtie \mathsf{G}$ *iff* $\mathsf{GF} \equiv \mathsf{FG}$ *in c-proportoids.*

Proof. The direction from right to left holds trivially. For the other direction, we have

$$\dfrac{\dfrac{\dfrac{\mathsf{F} \bowtie \mathsf{G}}{a : \mathsf{GF}a :: a : \mathsf{FG}a}}{a : a : \mathsf{GF}a : \mathsf{FG}a}}{\mathsf{GF} \equiv \mathsf{FG}.}\ \mathrm{c}$$

□

Fact 97. $\mathsf{F} \bowtie \mathsf{I}$ *holds for every mapping* F.

Proof. An immediate consequence of reflexivity (1). □

Fact 98. *The bowtie relation is reflexive and symmetric.*

Proof. Follows from the reflexivity (1) and symmetry (2) of the analogical proportion relation. □

Fact 99. *For any mapping* F*, we have* $\mathsf{F}^k \bowtie \mathsf{F}^\ell$ *for all* $k, \ell \geq 0$*. In particular, we have* $\mathsf{F} \bowtie \mathsf{F}^2$ *which means that* F *is idempotent with respect to bowtie equivalence.*

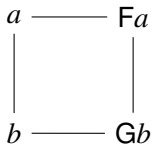
Proof. A direct consequence of reflexivity (1). □

Example 100. In $(\mathbb{N}, ::)$, we have $S^k \bowtie S^\ell$ for all $k, \ell \geq 0$.

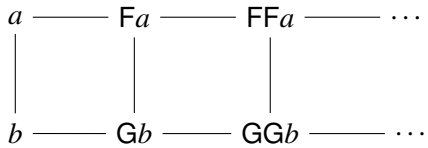
10.7. **Square equivalence.** Given $\mathsf{F}, \mathsf{G} : \mathfrak{P} \to \mathfrak{P}$, define their (***proportional***) ***square equivalence*** by

$$\mathsf{F} \square \mathsf{G} \quad :\Leftrightarrow \quad a : \mathsf{F}a :: b : \mathsf{G}b \quad \text{and} \quad a : b :: \mathsf{F}a : \mathsf{G}b, \quad \text{for all } a, b \in P,$$

which can be depicted as follows:



Applying F and G iteratively yields:



Notice the similarity between square equivalence and the definition of an analogy in §7, which immediately yields the following observation:

Fact 101. $\mathsf{F} \square \mathsf{F}$ *iff* F *is an analogy holds in any c-proportoid.*

Fact 102. *Proportional square equivalence is an equivalence relation for analogies on any t-proportoid.*

Proof. Reflexivity follows from the fact that every mapping is an analogy by assumption (and see Fact 101), symmetry follows from (2), and transitivity follows from (8). □

The next result shows that square equivalence is a very strong condition implying equivalence and join equivalence:

Fact 103. $\dfrac{\mathsf{F} \sqsubset \mathsf{G}}{\mathsf{F} \equiv \mathsf{G}}$.

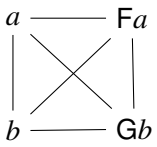
Fact 104. $\dfrac{\mathsf{F} \sqsubset \mathsf{G}}{\mathsf{F} \vee \mathsf{G}}$.

Example 105. As a direct consequence of Example 92 and Fact 104, in $(\mathbb{N}, ::)$ we have $S^k \sqsubset S^\ell$ iff $k = \ell$.

10.8. **Complete square equivalence.** We define the ***(proportional) complete square equivalence*** of F and G by

$$\mathsf{F} \boxtimes \mathsf{G} \quad :\Leftrightarrow \quad a : b :: \mathsf{F}a : \mathsf{G}b \quad \text{and} \quad a : \mathsf{F}a :: b : \mathsf{G}b \quad \text{and} \quad a : \mathsf{G}b :: b : \mathsf{F}a \quad \text{for all } a, b \in P.$$

This can be depicted as follows:



Fact 106. $\dfrac{\mathsf{F} \boxtimes \mathsf{G}}{\mathsf{F} \sqsubset \mathsf{G}}$.

Fact 107. $\dfrac{\mathsf{F} \boxtimes \mathsf{G}}{\mathsf{F} \equiv \mathsf{G}}$.

Proof.

$$\dfrac{\dfrac{\mathsf{F} \boxtimes \mathsf{G}}{\mathsf{F} \sqsubset \mathsf{G}}\,106}{\mathsf{F} \equiv \mathsf{G}.}\,103$$

□

11. Proportional circles

The following construction appears conceptually appealing:

Definition 108. Given $a, b \in P$, we define the ***(proportional) circle*** with center a and "radius" $\overline{ab}$ by the solution set

$$\mathcal{C}_{ab} := \mathcal{S}(a : b :: a : x).$$

The next result shows how to construct further solutions from a given one in any t-proportoid using circles:

Theorem 109. $\dfrac{d \in \mathcal{S}(a : b :: c : x)}{\mathcal{C}_{cd} \subseteq \mathcal{S}(a : b :: c : x)}\ t.$

Proof. We show the implication

$$e \in \mathcal{C}_{cd} \quad \Rightarrow \quad e \in \mathcal{S}(a : b :: c : x)$$

with the following derivation:

$$\dfrac{\dfrac{\dfrac{d \in \mathcal{S}(a : b :: c : x)}{a : b :: c : d} \qquad \dfrac{e \in \mathcal{C}_{cd}}{c : d :: c : e}}{a : b :: c : e}\,\mathrm{t}}{e \in \mathcal{S}(a : b :: c : x).}$$

□

12. Function proportions

Every relation on P can be extended point-wise to a relation on functions on P of same arity. For the analogical proportion relation extended to functions, we thus obtain the following definition:

Definition 110. Given mappings $\mathsf{E}, \mathsf{F}, \mathsf{G}, \mathsf{H} : \mathfrak{P} \to \mathfrak{P}$, we define the ***function proportion relation*** by

$$\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H} \quad :\Leftrightarrow \quad \mathsf{E}a : \mathsf{F}a :: \mathsf{G}a : \mathsf{H}a, \quad \text{for all } a \in P. \tag{25}$$

Fact 111. *We have*

$$\mathsf{E} : \mathsf{F} :: \mathsf{E} : \mathsf{F} \quad \textit{(reflexivity)},$$
$$\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H} \quad \Leftrightarrow \quad \mathsf{G} : \mathsf{H} :: \mathsf{E} : \mathsf{F} \quad \textit{(symmetry)},$$
$$\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H} \quad \Leftrightarrow \quad \mathsf{F} : \mathsf{E} :: \mathsf{H} : \mathsf{G} \quad \textit{(inner symmetry)}.$$

Similarly, all other properties of analogical proportions in Definition 2 transfer to function proportions.

Remark 112. Fact 111 means that from any proportoid $\mathfrak{P} = (P, ::_{\mathfrak{P}})$, we can construct the proportoid

$$\mathfrak{P}^{\mathfrak{P}} := (P^P, ::_{\mathfrak{P}^{\mathfrak{P}}})$$

of unary functions on P with $::_{\mathfrak{P}^{\mathfrak{P}}}$ defined point-wise as in (25).

Proposition 113. *In any c-proportoid, we have*

$$\mathsf{E} : \mathsf{I} :: \mathsf{F} : \mathsf{I} \quad \Leftrightarrow \quad \mathsf{E} \equiv \mathsf{F}.$$

Proof. We have

$$\begin{aligned} \mathsf{E} : \mathsf{I} :: \mathsf{F} : \mathsf{I} \quad &\Leftrightarrow \quad \mathsf{E}a : a :: \mathsf{F}a : a, \quad \text{for all } a \in P \\ &\overset{c}{\Leftrightarrow} \quad \mathsf{E}a : \mathsf{F}a :: a : a, \quad \text{for all } a \in P \\ &\overset{s}{\Leftrightarrow} \quad a : a :: \mathsf{E}a : \mathsf{F}a, \quad \text{for all } a \in P \\ &\Leftrightarrow \quad \mathsf{E} \equiv \mathsf{F}. \end{aligned}$$

□

The next result shows that proportional equivalence is compatible with function proportions:

Proposition 114. $\dfrac{\mathsf{E} \equiv \mathsf{F} \qquad \mathsf{G} \equiv \mathsf{H}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}}\ t.$

Proof.

$$\dfrac{\dfrac{\dfrac{\mathsf{E} \equiv \mathsf{F}}{a : a :: \mathsf{E}a : \mathsf{F}a}\ \mathrm{s}}{\mathsf{E}a : \mathsf{F}a :: a : a} \qquad \dfrac{\mathsf{G} \equiv \mathsf{H}}{a : a :: \mathsf{G}a : \mathsf{H}a}\ \mathrm{t}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}.}$$

□

Corollary 115. $\dfrac{\mathsf{E} \sqsubset \mathsf{F} \qquad \mathsf{G} \sqsubset \mathsf{H}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}}\ t.$

Proof.

$$\dfrac{\dfrac{\mathsf{E} \sqsubset \mathsf{F}}{\mathsf{E} \equiv \mathsf{F}}\ 103 \qquad \dfrac{\mathsf{G} \sqsubset \mathsf{H}}{\mathsf{G} \equiv \mathsf{H}}\ 103}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}.}\ \mathrm{t},\ \text{Proposition 114}$$

□

Corollary 116. $\dfrac{\mathsf{E} \boxtimes \mathsf{F} \qquad \mathsf{G} \boxtimes \mathsf{H}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}}\ t.$

Proof.

$$\dfrac{\dfrac{\mathsf{E} \boxtimes \mathsf{F}}{\mathsf{E} \sqsubset \mathsf{F}}\ 106 \qquad \dfrac{\mathsf{G} \boxtimes \mathsf{H}}{\mathsf{G} \sqsubset \mathsf{H}}\ 106}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}.}\ \mathrm{t},\ \text{Corollary 115}$$

□

Corollary 117. $\dfrac{\mathsf{E} \triangledown \mathsf{F} \qquad \mathsf{G} \triangledown \mathsf{H}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}}\ ct.$

Proof.

$$\dfrac{\dfrac{\mathsf{E} \triangledown \mathsf{F}}{\mathsf{E} \equiv \mathsf{F}}\ \mathrm{c},\ \text{Proposition 94} \qquad \dfrac{\mathsf{G} \triangledown \mathsf{H}}{\mathsf{G} \equiv \mathsf{H}}\ \mathrm{c},\ \text{Proposition 94}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}.}\ \mathrm{t},\ \text{Proposition 114}$$

□

Definition 118. We extend proportional identity from elements of P to unary mappings on P point-wise by

$$\mathsf{F} \ddot{=} \mathsf{G} \quad :\Leftrightarrow \quad \mathsf{F}a \ddot{=} \mathsf{G}a, \quad \text{for all } a \in P.$$

The next result is a generalization of Theorem 61 from elements to functions:

Proposition 119. $\dfrac{\mathsf{E} \ddot{=} \mathsf{F} \qquad \mathsf{G} \ddot{=} \mathsf{H}}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}}\ t.$

Proof.

$$\dfrac{\dfrac{\dfrac{\mathsf{E} \ddot{=} \mathsf{F}}{\mathsf{E}a \ddot{=} \mathsf{F}a} \qquad \dfrac{\mathsf{G} \ddot{=} \mathsf{H}}{\mathsf{G}a \ddot{=} \mathsf{H}a}}{\mathsf{E}a : \mathsf{F}a :: \mathsf{G}a : \mathsf{H}a}\ 61}{\mathsf{E} : \mathsf{F} :: \mathsf{G} : \mathsf{H}.}$$

□

13. Proportional similarity

In this section, we shall introduce a notion of similarity in terms of analogical proportions:

Definition 120. Given $a \in P$ and $b \in R$ and some set of functions $\Sigma := \{\sigma_{ab} : P \to R \mid a, b \in P\}$, define

$$a \lesssim_\Sigma b \quad :\Leftrightarrow \quad a : c :: b : \sigma_{ab}c, \quad \text{for every } c \in P,$$

and

$$a \approx_\Sigma b \quad :\Leftrightarrow \quad a \lesssim_\Sigma b \quad \text{and} \quad b \lesssim_\Sigma a.$$

In case $a \approx_\Sigma b$, we say that a and b are **Σ-*similar***.

Notice that

$$\begin{aligned} a \lesssim_\Sigma b \quad &\Leftrightarrow \quad \text{for every } c \in P \text{ there is some } d = \sigma_{ab}c \in R \text{ such that } a : c :: b : d \\ &\Rightarrow \quad \mathcal{S}(a : c :: b : x) \neq \emptyset, \quad \text{for all } c \in P. \end{aligned}$$

Proposition 121. *Σ-similarity is reflexive and symmetric in any proportoid. Moreover, if*

$$\sigma_{ab}\sigma_{ca}d = \sigma_{cb}d \tag{26}$$

holds for all $a, b, c, d \in P$, then Σ-similarity is an equivalence relation in any t-proportoid.

Proof. Reflexivity follows from (1) which guarantees that for any $c \in P$ there is some $d := c \in P$ such that $a : c :: a : c$. Symmetry holds trivially. To prove transitivity, we proceed as follows. Suppose $a \approx_\Sigma b$ and $b \approx_\Sigma c$, which means that

$$\begin{aligned} a : d :: b : \sigma_{ab}d, \quad &\text{for all } d \in P, \\ b : e :: c : \sigma_{bc}e, \quad &\text{for all } e \in P. \end{aligned}$$

Let $f \in P$ be an arbitrary element. We then have the following derivation:

$$\dfrac{\dfrac{\dfrac{a : f :: b : \sigma_{ab}f \qquad b : \sigma_{ab}f :: c : \sigma_{bc}\sigma_{ab}f}{a : f :: c : \sigma_{bc}\sigma_{ab}f}\,\mathrm{t}}{a : f :: c : \sigma_{cb}d}\,26}{a \approx_\Sigma c.}$$

□

Theorem 122. *Let $\mathfrak{P} = (P, ::)$ be an it-proportoid. If there exists a set of functions Σ satisfying* (26) *and the proportions*

$$\sigma_{ae}b : \sigma_{bf}a :: \sigma_{cg}d : \sigma_{dh}c \tag{27}$$

for all $a, b, c, d, e, f, g, h \in P$, then $\approx_\Sigma$ is a proportional congruence thus satisfying

$$\frac{a \approx_\Sigma e \quad b \approx_\Sigma f \quad c \approx_\Sigma g \quad d \approx_\Sigma h \qquad a : b :: c : d}{e : f :: g : h}.$$

Proof. We have the derivations

$$\dfrac{\dfrac{\dfrac{\dfrac{a \approx_\Sigma e}{a : b :: e : \sigma_{ae}b}\,\mathrm{s}}{e : \sigma_{ae}b :: a : b} \qquad a : b :: c : d}{e : \sigma_{ae}b :: c : d}\,\mathrm{t} \qquad \dfrac{c \approx_\Sigma g}{c : d :: g : \sigma_{cg}d}\,\mathrm{t}}{e : \sigma_{ae}b :: g : \sigma_{cg}d}\,\mathrm{t}$$

and

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{b \approx_\Sigma f}{b : a :: f : \sigma_{bf}a}\,\mathrm{y}}{a : b :: \sigma_{bf}a : f}\,\mathrm{s}}{\sigma_{bf}a : f :: a : b} \qquad a : b :: c : d}{\sigma_{bf}a : f :: c : d}\,\mathrm{t} \qquad \dfrac{\dfrac{d \approx_\Sigma h}{d : c :: h : \sigma_{dh}c}\,\mathrm{y}}{c : d :: \sigma_{dh}c : h}\,\mathrm{y}}{\sigma_{bf}a : f :: \sigma_{dh}c : h.}\,\mathrm{t}$$

Now since we assume (27) and inner transitivity, we have

$$\dfrac{\dfrac{e : \sigma_{ae}b :: g : \sigma_{cg}d \qquad \sigma_{ae}b : \sigma_{bf}a :: \sigma_{cg}d : \sigma_{dh}c}{e : \sigma_{bf}a :: g : \sigma_{dh}c}\,\mathrm{i} \qquad \sigma_{bf}a : f :: \sigma_{dh}c : h}{e : f :: g : h.}\,\mathrm{i}$$

□

Fact 123. *For any analogy A and any element a, we have $a \lesssim_\Sigma \mathsf{A}a$.*

14. Proportoids in universal algebra and predicate logic

This paper is axiomatic in style in the sense that we do not study concrete realizations of the analogical proportion relation. However, in a series of papers the author has shown how a *canonical* notion of an analogical proportion relation can be constructed from any algebra or structure in the sense of universal algebra and first-order logic, respectively [1, 4].

More formally, given a first-order language L consisting of ranked function[5] and relation symbols, an ***L-structure*** consists of a non-empty set P together with concrete functions and relations on P corresponding to the function and relation symbols in L (see e.g. [9, §2]).

Given an L-structure $\mathfrak{A} = (A, \mathcal{F}, \mathcal{R})$ with functions $\mathcal{F}$ and relations $\mathcal{R}$, we can define the ***analogical proportion relation*** in $\mathfrak{A}$ — in symbols,

$$a : b ::_{\mathfrak{A}} c : d$$

as in [4]. We do not want to go into technical details here. What is of interest here is that we can *canonically* associate with any such L-structure $\mathfrak{A} = (A, \mathcal{F}, \mathcal{R})$ a proportoid $\mathfrak{A}^{::} := (A, ::_{\mathfrak{A}})$ by defining $::_{\mathfrak{A}}$ as described in [4].

We can use algebras and structures to represent proportoids:

Definition 124. Let L be a set of function symbols. Then we call an L-algebra $\mathfrak{A} = (A, L^{\mathfrak{A}})$ a ***representation*** of a proportoid $(A, ::)$ iff there are 4-ary L-terms s, t such that

$$a : b :: c : d \quad \Leftrightarrow \quad s^{\mathfrak{A}}(a, b, c, d) = t^{\mathfrak{A}}(a, b, c, d)$$

holds for all $a, b, c, d \in A$. A proportoid is ***representable*** iff it has a representation.

Example 125. Every metric proportoid $M = (\mathbb{R}, ::_{\delta})$ arising from a metric space $(\mathbb{R}, \delta)$ is, by definition (13), representable.

15. Conclusion

This paper introduced proportoids as sets endowed with a 4-ary analogical proportion relation satisfying a suitable set of axioms which are rooted in but different from Lepage's initial axiomatization. We then introduced proportional homomorphisms and their congruences and showed that they are related in the usual sense via kernels. Moreover, we introduced proportional analogies and showed that in transitive proportoids, they satisfy the strong proportion-preserving principle and are thus closely related to proportional homomorphisms. We showed how partial proportional analogies can be constructed from an enumeration and a selection function. We introduced a number of binary relations between unary mappings on proportoids. Finally, we introduced a notion of similarity in terms of analogical proportions.

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[5] Constant symbols are omitted and identified with 0-ary function symbols.